Activities

Overview

Analyzing questions related to OSPO activities and services.

Research questions:
- What services do OSPOs most commonly offer?
- How do OSPOs spend the bulk of their time?

Survey questions:
- Tick the activities your OSPO actually carries out today:
- Developing and sharing best practices (best_practices)
- Educational efforts (edu_efforts)
- External partnerships and tech transfer (ext_partnerships)
- [Supporting tools and infrastructure (tools_infra) - not analyzed here, see tools notebook]
- Community building (comm_building)
- Directly supporting open source projects (project_support)
- Advocacy and policy (advocacy_policy)
- Funding and financial support (funding_support)
- OSPO operations (ospo_ops)
- Of the activities above, which take the most of your own time? Rank your top three. (ranking)

Import packages, utilities, and config variables

```
library(readxl)
library(dplyr)
```

```
Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
library(tibble)
```

```
Warning: package 'tibble' was built under R version 4.4.3
```

```
library(docstring)
```

```

Attaching package: 'docstring'

The following object is masked from 'package:utils':

    ?
```

```
library(readr)
```

```
Warning: package 'readr' was built under R version 4.4.3
```

```
library(tidyr)
```

```
Warning: package 'tidyr' was built under R version 4.4.3
```

```
source("utils.R")

# Read config.yml
cfg <- config::get()
```

Load data

```
org_responses <- read_excel(file.path(cfg$curated_path, "orgs.xlsx"), sheet="responses")
codebook <- read_excel(file.path(cfg$data_path, "codebook.xlsx"))
best_practices <- read_tsv(file.path(cfg$data_path, "best_practices.tsv"), show_col_types = I
edu_efforts <- read_tsv(file.path(cfg$data_path, "edu_efforts.tsv"), show_col_types = FALSE,
ext_partnerships <- read_tsv(file.path(cfg$data_path, "ext_partnerships.tsv"), show_col_types
tools_infra <- read_tsv(file.path(cfg$data_path, "tools_infra.tsv"), show_col_types = FALSE,
comm_building <- read_tsv(file.path(cfg$data_path, "comm_building.tsv"), show_col_types = FA
project_support <- read_tsv(file.path(cfg$data_path, "project_support.tsv"), show_col_types =
advocacy_policy <- read_tsv(file.path(cfg$data_path, "advocacy_policy.tsv"), show_col_types =
funding_support <- read_tsv(file.path(cfg$data_path, "funding_support.tsv"), show_col_types =
ospo_ops <- read_tsv(file.path(cfg$data_path, "ospo_ops.tsv"), show_col_types = FALSE, skip_e
ranking <- read_tsv(file.path(cfg$data_path, "ranking.tsv"), show_col_types = FALSE, skip_emp
sata_other <- read_tsv(file.path(cfg$data_path, "sata_other.tsv"), show_col_types = FALSE, sk
```

What services do OSPOs most commonly offer?

```
best_practices$organization <- org_responses$organization
bp_consolidated <- consolidate_org_responses_yn(best_practices)
bp <- count_yes_responses(bp_consolidated)
bp
```

```
# A tibble: 6 x 3
  count tip                                                              nickname
  <dbl> <chr>                                                            <chr>
1    18 Producing guides, templates, or checklists for open source work guides_~
2    16 Curating or adapting external resources for your institutional~ externa~
3    14 Contributing practices or patterns back to external communitie~ externa~
4    13 Maintaining an internal knowledge base, wiki, or website of pr~ knowled~
5    12 Documenting internal case studies or success stories            case_st~
6    10 Providing AI services or infrastructure for open source projec~ ai_serv~
```

```
edu_efforts$organization <- org_responses$organization
edu_consolidated <- consolidate_org_responses_yn(edu_efforts)
edu <- count_yes_responses(edu_consolidated)
edu
```

```
# A tibble: 5 x 3
  count tip                                                              nickname
```

```
  <dbl> <chr>                                                          <chr>
1    20 Running training sessions or workshops on open source topics  worksho~
2    19 Mentoring individuals or project teams                        mentori~
3    13 Running or supporting student open source programs (internship~ student~
4    10 Supporting open source content or projects in the curriculum   curricu~
5     8 Providing onboarding materials for newcomers to open source    onboard~
```

```
ext_partnerships$organization <- org_responses$organization
ext_consolidated <- consolidate_org_responses_yn(ext_partnerships)
ext <- count_yes_responses(ext_consolidated)
ext
```

```
# A tibble: 5 x 3
  count tip                                                          nickname
  <dbl> <chr>                                                        <chr>
1    24 Advising on open source licensing for research outputs       licensi~
2    23 Representing the institution in external open source communiti~ represe~
3    18 Working with the technology transfer or knowledge exchange off~ tto_rel~
4    18 Building partnerships with companies, foundations, or other ex~ partner~
5    14 Supporting commercialization pathways for open source projects~ commerc~
```

```
tools_infra$organization <- org_responses$organization
tools_consolidated <- consolidate_org_responses_yn(tools_infra)
tools <- count_yes_responses(tools_consolidated)
tools
```

```
# A tibble: 8 x 3
  count tip                                                          nickname
  <dbl> <chr>                                                        <chr>
1    15 Maintaining a catalog or registry of the institution's open so~ catalog
2    14 Providing or managing code hosting (for example an institution~ code_ho~
3     9 <NA>                                                          metrics
4     9 Providing metrics or dashboards on open source activity       metrics
5     8 Automating checks for licensing, security, or good practice   checks_~
6     8 Building Open Source software to support research             build_r~
7     7 Providing development infrastructure such as CI/CD or testing ~ dev_inf~
8     4 Building other Open Source software (e.g. EdTech)            build_o~
```

```
comm_building$organization <- org_responses$organization
cb_consolidated <- consolidate_org_responses_yn(comm_building)
cb <- count_yes_responses(cb_consolidated)
cb
```

```
# A tibble: 5 x 3
  count tip                                                          nickname
  <dbl> <chr>                                                        <chr>
1    19 Organizing events such as conferences, hackathons, or showcases events
2    16 Connecting people across departments who are working on simila~ cross_d~
3    12 Running a community of practice, regular meetups, or discussio~ cop_mee~
4     8 Publishing newsletters or internal communications that tell op~ communi~
5     3 Running recognition or reward mechanisms for open source contr~ recogni~
```

```
project_support$organization <- org_responses$organization
ps_consolidated <- consolidate_org_responses_yn(project_support)
ps <- count_yes_responses(ps_consolidated)
ps
```

```
# A tibble: 5 x 3
  count tip                                                          nickname
  <dbl> <chr>                                                        <chr>
1    21 Advising projects on open-sourcing research or internal code    advisin~
2    16 Helping projects set up governance and contribution processes   governa~
3    14 Providing hands-on help, such as repository reviews or documen~ hands_on
4     7 Assessing project or community health and contributor readiness health_~
5     5 Incubating or hosting projects within the OSPO                  incubat~
```

```
advocacy_policy$organization <- org_responses$organization
ap_consolidated <- consolidate_org_responses_yn(advocacy_policy)
ap <- count_yes_responses(ap_consolidated)
ap
```

```
# A tibble: 6 x 3
  count tip                                                          nickname
  <dbl> <chr>                                                        <chr>
1    18 Briefing senior leadership on the value of open source          brief_l~
2    17 Advocating for open source within institutional strategy proce~ strateg~
3    15 Developing or advising on institutional open source policies    policy_~
4    13 Working on addressing open source in AI policy                  ai_poli~
5    11 Working with legal, compliance, or security teams to remove bl~ legal_c~
6    10 Engaging with national, funder, or sector policy developments   externa~
```

```
funding_support$organization <- org_responses$organization
fs_consolidated <- consolidate_org_responses_yn(funding_support)
fs <- count_yes_responses(fs_consolidated)
fs
```

```
# A tibble: 6 x 3
  count tip                                                      nickname
  <dbl> <chr>                                                    <chr>
1    19 Supporting grant applications and grant writing         grant_suppo~
2    13 Helping projects identify external funding sources      ext_funding~
3    11 Fundraising to sustain the OSPO function itself         ospo_fundra~
4     9 Helping projects develop sustainability or funding models sustainabil~
5     6 Administering internal grants or funds for open source work internal_gr~
6     5 Directly funding open source projects                    direct_fund~
```

```
ospo_ops$organization <- org_responses$organization
ospo_consolidated <- consolidate_org_responses_yn(ospo_ops)
ospo <- count_yes_responses(ospo_consolidated)
ospo
```

```
# A tibble: 5 x 3
  count tip                                                      nickname
  <dbl> <chr>                                                    <chr>
1    21 Reporting on OSPO activities or impact to leadership, funders,~ reporti~
2    21 Developing or maintaining the OSPO strategy or roadmap  strateg~
3    19 Day-to-day operations: administration, budgeting, and internal~ day_to_~
4    13 Hiring, managing or developing OSPO staff, students or interns staffing
5    13 Coordinating volunteers, champions, or contributors beyond the~ volunte~
```

Let's look at the upper quartile of activities, sorted by how widespread they are among OSPOs.

```
mega_df <- rbind(bp, edu, ext, tools, cb, ps, ap, fs, ospo)
top_quartile <- mega_df |>
  filter(count >= quantile(count, 0.75)) |>
  arrange(desc(count))
top_quartile
```

```
# A tibble: 14 x 3
   count tip                                                     nickname
   <dbl> <chr>                                                   <chr>
 1    24 Advising on open source licensing for research outputs  licensi~
 2    23 Representing the institution in external open source communit~ represe~
 3    21 Advising projects on open-sourcing research or internal code advisin~
 4    21 Reporting on OSPO activities or impact to leadership, funders~ reporti~
 5    21 Developing or maintaining the OSPO strategy or roadmap  strateg~
```

```
 6    20 Running training sessions or workshops on open source topics   worksho~
 7    19 Mentoring individuals or project teams                         mentori~
 8    19 Organizing events such as conferences, hackathons, or showcas~ events
 9    19 Supporting grant applications and grant writing                grant_s~
10    19 Day-to-day operations: administration, budgeting, and interna~ day_to_~
11    18 Producing guides, templates, or checklists for open source wo~ guides_~
12    18 Working with the technology transfer or knowledge exchange of~ tto_rel~
13    18 Building partnerships with companies, foundations, or other e~ partner~
14    18 Briefing senior leadership on the value of open source         brief_l~
```

Maybe it would be interesting to look at the least widespread activities as well.

```
mega_df |> filter(count <= quantile(count, 0.25)) |> arrange(count)
```

```
# A tibble: 14 x 3
   count tip                                                          nickname
   <dbl> <chr>                                                        <chr>
 1     3 Running recognition or reward mechanisms for open source cont~ recogni~
 2     4 Building other Open Source software (e.g. EdTech)              build_o~
 3     5 Incubating or hosting projects within the OSPO                 incubat~
 4     5 Directly funding open source projects                          direct_~
 5     6 Administering internal grants or funds for open source work    interna~
 6     7 Providing development infrastructure such as CI/CD or testing~ dev_inf~
 7     7 Assessing project or community health and contributor readine~ health_~
 8     8 Providing onboarding materials for newcomers to open source    onboard~
 9     8 Automating checks for licensing, security, or good practice    checks_~
10     8 Building Open Source software to support research              build_r~
11     8 Publishing newsletters or internal communications that tell o~ communi~
12     9 <NA>                                                           metrics
13     9 Providing metrics or dashboards on open source activity        metrics
14     9 Helping projects develop sustainability or funding models      sustain~
```

That is pretty interesting! OSPOs are too poor for these activities lol.

Which activities take the most time?

Let's say that every time an activity is ranked #1, it gets 3 points, when it's ranked #2, it gets 2 point, and when it's ranked #3, it gets 1 point.

I think in this case I will not filter to one response per organization, since the survey said, "Answer for yourself, not the whole team." So we can interpret this as, how OSPO workers spend their time.

```
activity_scores <- ranking |>
  pivot_longer(
    cols = c(rank1, rank2, rank3),
    names_to = "rank",
    values_to = "activity"
  ) |>
  mutate(
    points = case_when(
      rank == "rank1" ~ 3,
      rank == "rank2" ~ 2,
      rank == "rank3" ~ 1
    )
  ) |>
  group_by(activity) |>
  summarise(total_score = sum(points), .groups = "drop") |>
  arrange(desc(total_score))

activity_scores
```

```
# A tibble: 11 x 2
   activity                                    total_score
   <chr>                                             <dbl>
 1 OSPO operations and strategy                         35
 2 Community building                                   31
 3 External partnerships and tech transfer              31
 4 Directly supporting open source projects             23
 5 Educational efforts                                  21
 6 Advocacy and policy                                  14
 7 Developing and sharing best practices                13
 8 Supporting tools and infrastructure                  13
 9 Funding and financial support                        10
10 <NA>                                                  6
11 Other activities identified above                     1
```

Note: from manual inspection I see that one person did not answer this question. That's where the NAs come from.

Note: I manually reviewed the “other” answers and didn’t see anything super compelling. I did add a couple of “lessons learned” based on answers that people wrote in and that I think should be survey options next time.